

# Equivalent Ratios

Lesson 3-4

**Name:** Type your name      **Date:** Today's date      **Class:** Class period

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## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can find and check equivalent ratios using multiplication and division.

**Language Objective:** I can explain my reasoning using the words equivalent ratios, rate, proportion, and simplify.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Ratio

Equivalent ratios

Rate

Proportion

Simplify



Tap any word to see what it means and a picture.

1

A comparison of two quantities is a .

2

— Two ratios that mean the same thing.

3

A ratio that compares two quantities with different units, like miles per hour, is a

.

4

An equation showing that two ratios are equal is a .

5

Writing a ratio in its smallest equal form, like  $6:9 = 2:3$ , is to

it.

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### JUSTIFY

**Looking at the tomatoes-to-broth table (3:2, 6:4, 9:6, 12:8), how can you PROVE these ratios are all equivalent?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

### 2. Plan Your Math Words

Check at least two words you will use.

☐ **Equivalent ratios** — Two ratios that mean the same thing.  
*Razones equivalentes — Dos razones que significan lo mismo.*

☐ **Rate** — A ratio comparing two amounts with different units, like miles per hour.  
*Tasa — Una razón que compara dos cantidades con unidades distintas, como millas por hora.*

☐ **Proportion** — A math sentence saying two ratios are equal.  
*Proporción — Una oración matemática que dice que dos razones son iguales.*

☐ **Simplify** — To make a ratio smaller while keeping the same comparison.  
*Simplificar — Hacer una razón más pequeña sin cambiar la comparación.*

☐ **Ratio** — A way to compare two amounts.  
*Razón — Una manera de comparar dos cantidades.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**These ratios are equivalent because each is \_\_\_\_ times \_\_\_\_.**

*Estas razones son equivalentes porque cada una es \_\_\_\_ veces \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** 32 is 4 times 8, so she multiplies both ingredients by 4.

**Evidence:** A strong answer finds the scale factor 4 ( $32 \div 8$ ) and explains both ingredients must be multiplied by 4 to make equivalent ratios.

**Reasoning:** This evidence connects the definition of ratio to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **These ratios are equivalent because each is \_\_\_\_ times \_\_\_\_.**

*Estas razones son equivalentes porque cada una es \_\_\_\_ veces \_\_\_\_.*

- **I checked by \_\_\_\_ both numbers.**

*Lo comprobé al \_\_\_\_ ambos números.*

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*